

ML-based Ranking of ABO_3 perovskites for high specific capacitance

Presented to:

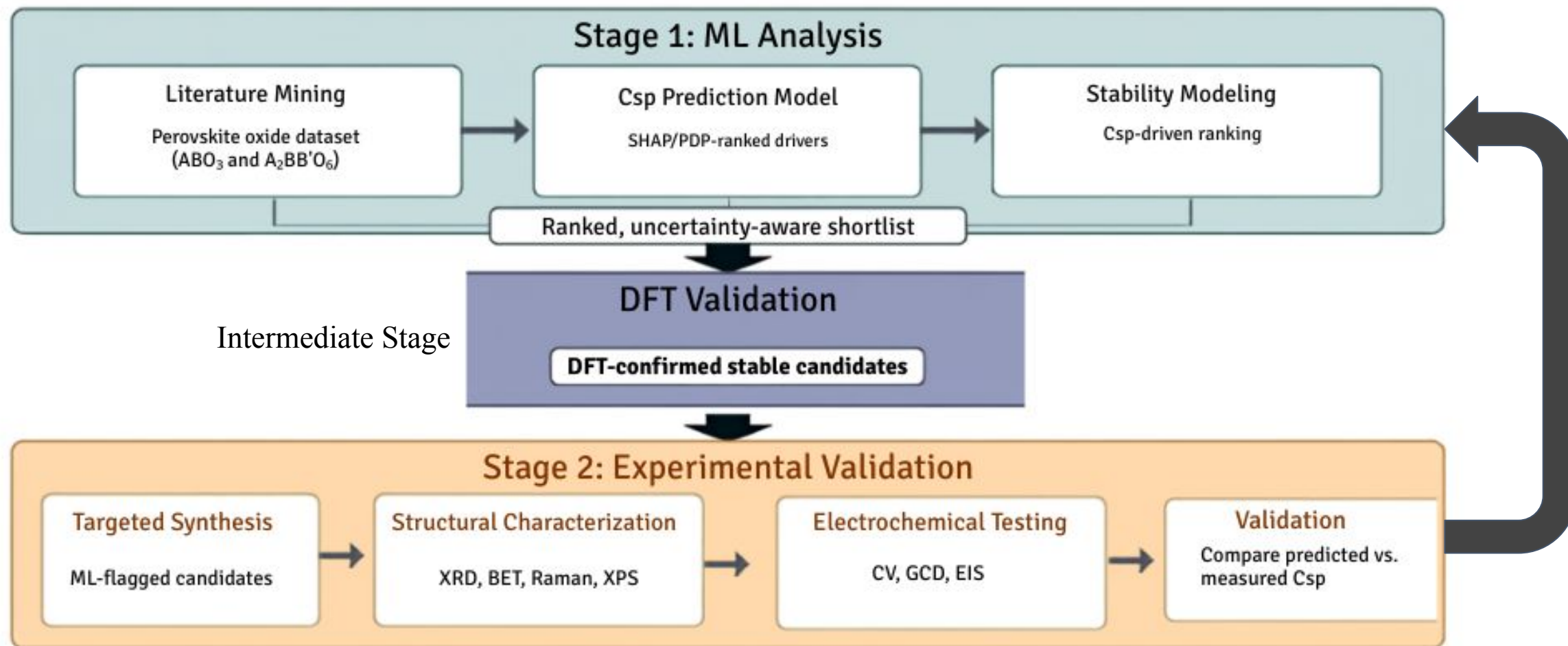
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This slide deck was made to present our ML approaches while it was still in continuous progress, so please do not expect it to be a final revised edition of our work in any means.

The Complete Project at a Glance: The Computation-guided Closed-Loop Pipeline



Where our two attempts landed

First Try

Regress C_{top}

Predict the exact F_{top} value.

Under paper-grouped CV: $R^2 \approx 0$
— no better than the mean.

Failed

Latest Try

Classify the top tier

Predict whether a composition
reaches top-33% C_{top} . Signal
recovered: ROC-AUC 0.806.

Current approach

Same data, same features, same models — the gain came from asking a question the data could answer confidently.

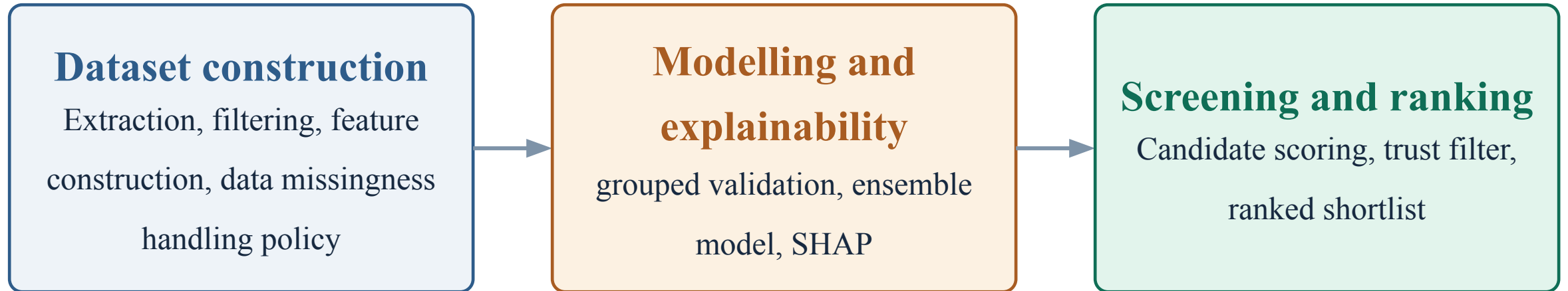
Why literature-mined $C_{\square\square}$ resists direct prediction

Four structural problems that dictate every design choice that follows

Problem	What it means	Our solution
Multi-source	46 paper IDs / 42 unique DOIs. Each lab has different fabrication and testing fingerprint.	Group by paper/DOI in CV
Condition heterogeneity	$C_{\square\square}$ depends on current density, electrolyte, loading — values are not comparable as reported.	Keep as features when train; fix when scoring/candidate ranking
Small n	189 usable GCD rows after filtering — far below what accurate regression needs.	Predict a tier, not a value
Reporting bias	High-performance results are over-reported; structural metadata often missing.	Explicit missingness flags

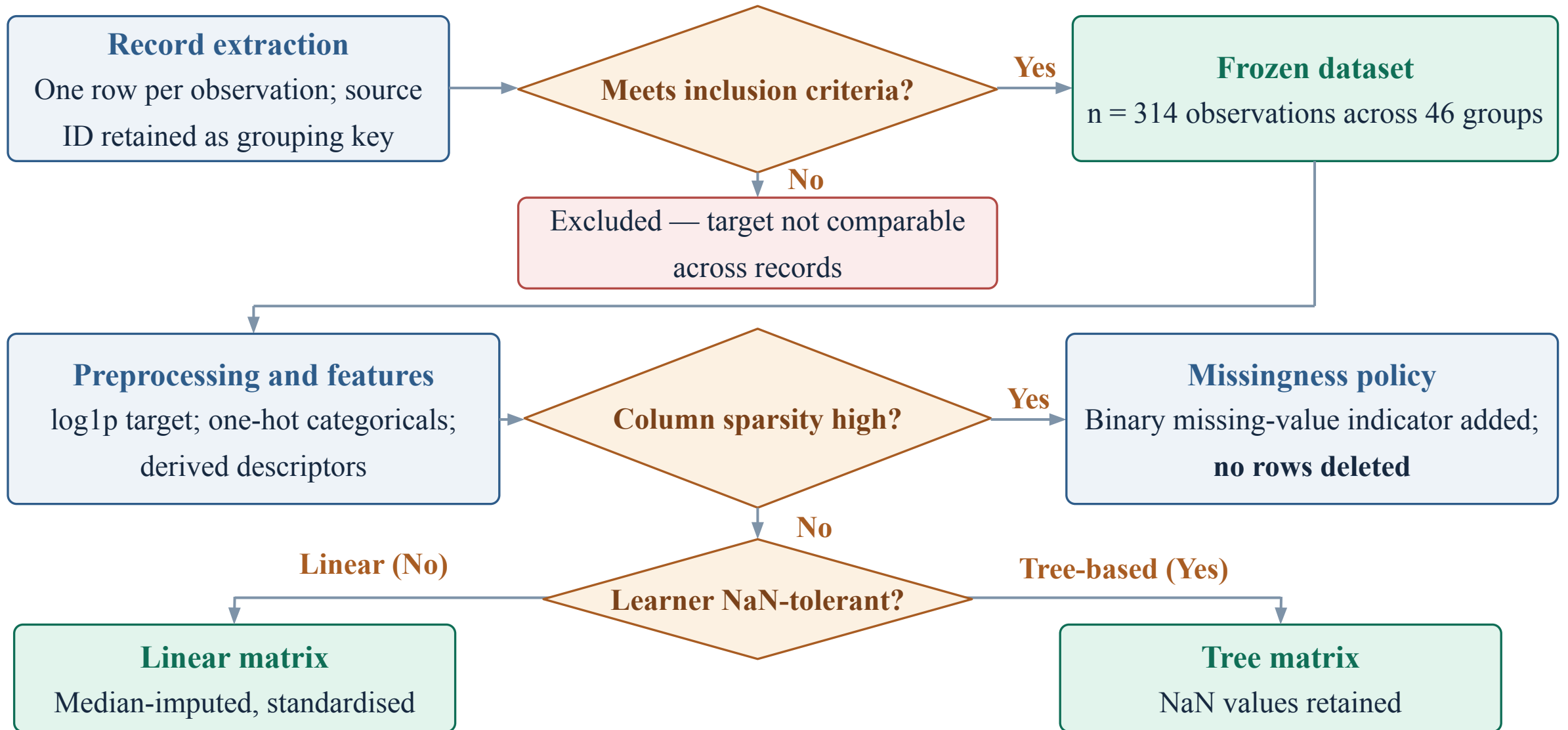
Overall ML workflow

Three stages



Dataset Preparation & Preprocessing techniques

Flowchart 1 — Dataset construction and preprocessing



The dataset

Frozen, literature-mined perovskite supercapacitor records

314 rows

Frozen dataset — 46 paper IDs, 42 unique DOIs

257 rows

ABO₃ subset — 38 papers

189 rows

GCD only — 29 papers
← modelled

63 / 126

Top-33% positives vs rest, from 18 papers

Feature families (79 after encoding)

Composition

A/B one-hots, dopant, fraction x

Structural chemistry

ionic radii, r_a/r_b , tolerance & octahedral factors, mixing entropy

Magpie statistics

mean/SD of melting T, electronegativity, valence, covalent radius

Synthesis

method, calcination T, morphology, crystal phase, crystallite size, BET

Test conditions

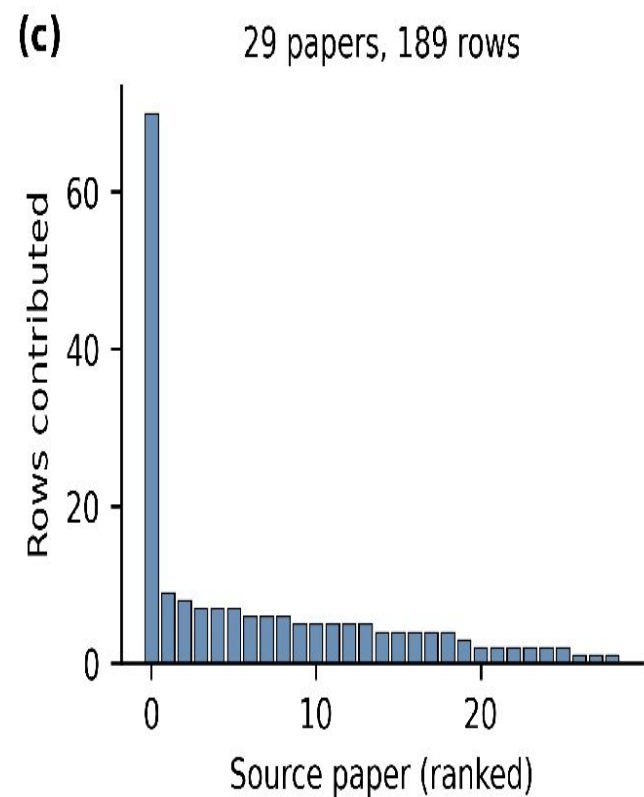
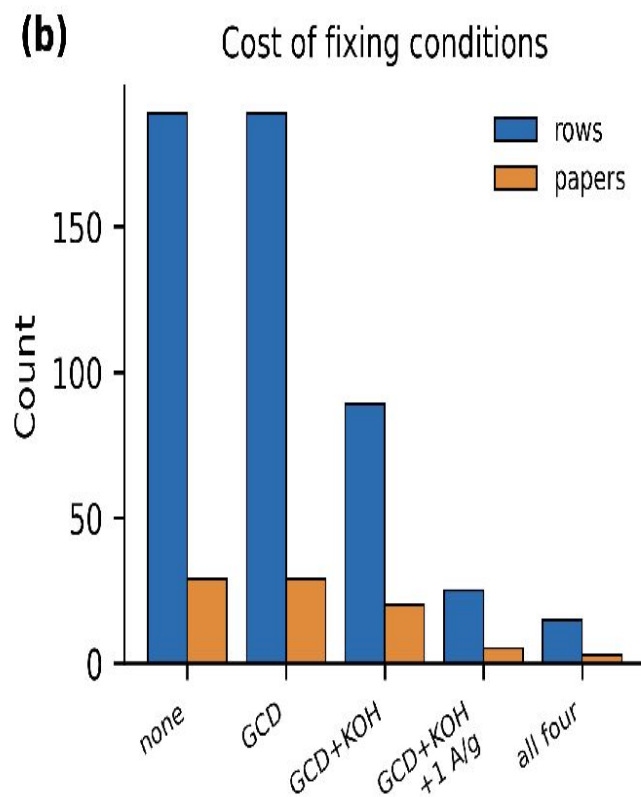
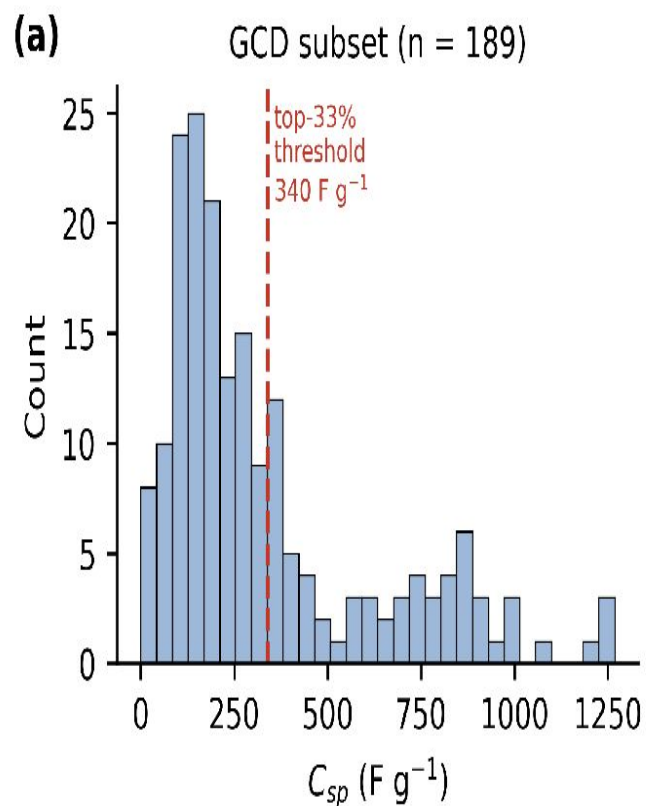
electrolyte, concentration, current density, scan rate, loading

Missingness

explicit *_missing flags — absence is informative

The label: top-33% of the GCD subset

Panel (a) distribution and threshold · (b) cost of fixing conditions · (c) rows per paper



Why a tier?

189 rows cannot support point prediction of an exact $F g^{-1}$ value.

Why 33%?

Separating a genuine top third is easier than splitting at the middle.

Not one lab

Positives span 18 of 29 papers — no single-source artefact.

The cost of fixing conditions while training

Fixing conditions at training time destroys the dataset; fixing them at scoring time does not

If we filter to $1 \text{ A g}^{-1} + \text{KOH} + \text{GCD} + 1 \text{ M}$

15 rows / 3 papers

Unusable. The screening condition cannot be a training filter.

If we filter to GCD only

189 rows / 29 papers

As CV- and GCD-derived $\text{C}\square\square$ are not comparable.

The resolution: filter vs. feature vs. query

FILTER

Measurement type = GCD

Defines what $\text{C}\square\square$ even means comparably

FEATURE

Electrolyte, concentration, current density, scan rate

Model learns their effect from all 189 rows

QUERY

Set to KOH, 1 M, 1 Ag^{-1} when scoring candidates

Level playing field; only chemistry varies

Not a single row were discarded unnecessarily.

The validation protocol

1 StratifiedGroupKFold

5 folds, grouped by source paper. No paper appears in both train and test — the model is always judged on an unseen lab. Class balance preserved per fold.

2 ROC-AUC & PR-AUC

Accuracy is misleading at 33/67 imbalance. So, kept No-skill (lucky) floors at: 0.500 (ROC) and 0.333 (PR) — quoted on every chart.

3 Label: top 33%

$C_{\text{top}} > 339.7 \text{ F g}^{-1}$ (67th percentile of the GCD subset).
Checked at 45% too: signal holds (ROC 0.68–0.80).

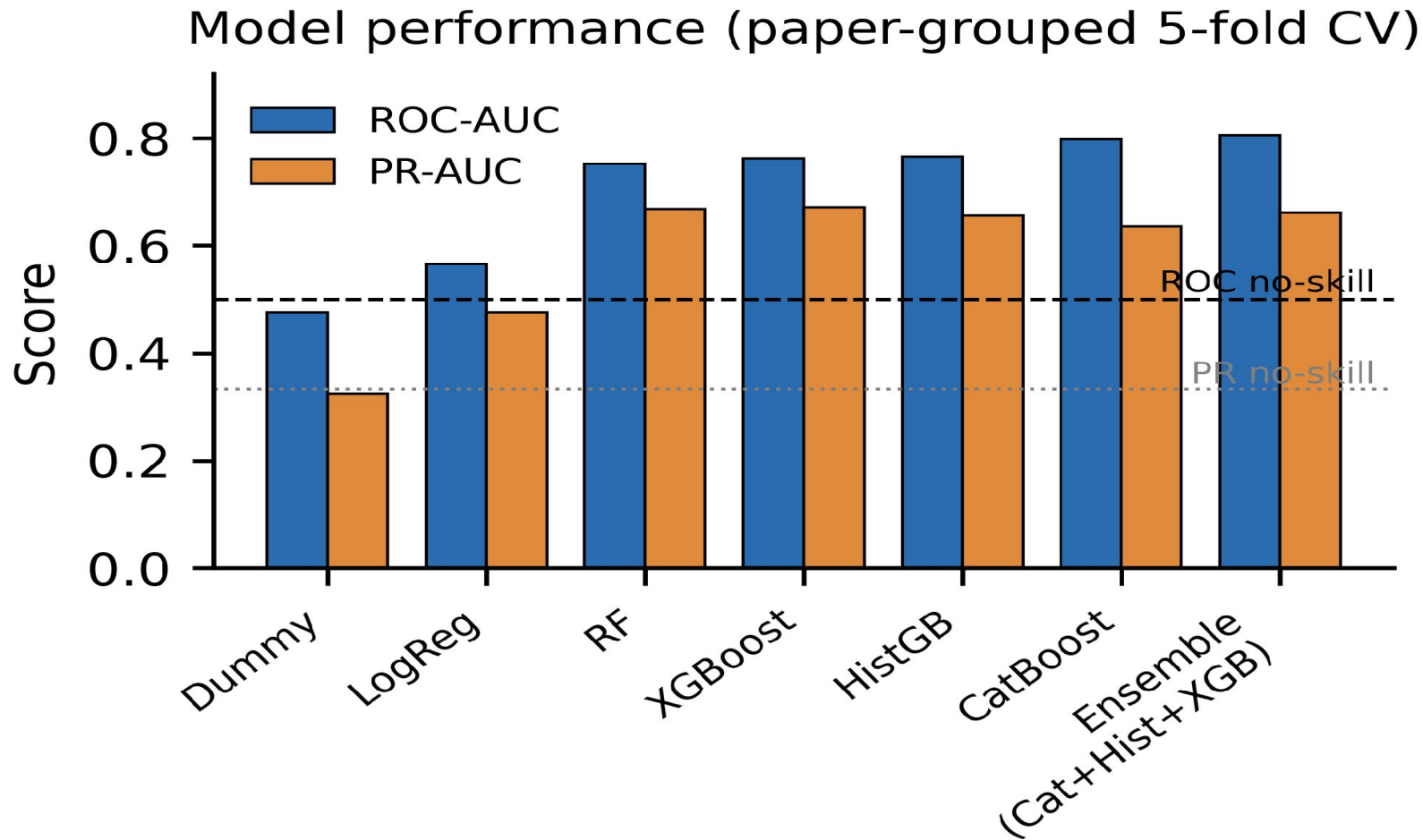
4 Fixed seeds

random_state = 0 throughout; tuned hyperparameters recorded for exact reproduction.

Grouped CV is deliberately [conservative](#). It is [why our numbers look lower than published alternatives](#) — and why this is defensible in the review bench comfortably.

Classification recovers the signal regression could not

All tree models clear the no-skill floor; logistic regression stays near chance



Regression failed

Grouped $R^2 \approx 0$ — no better than predicting the mean.

LogReg $\approx$ chance

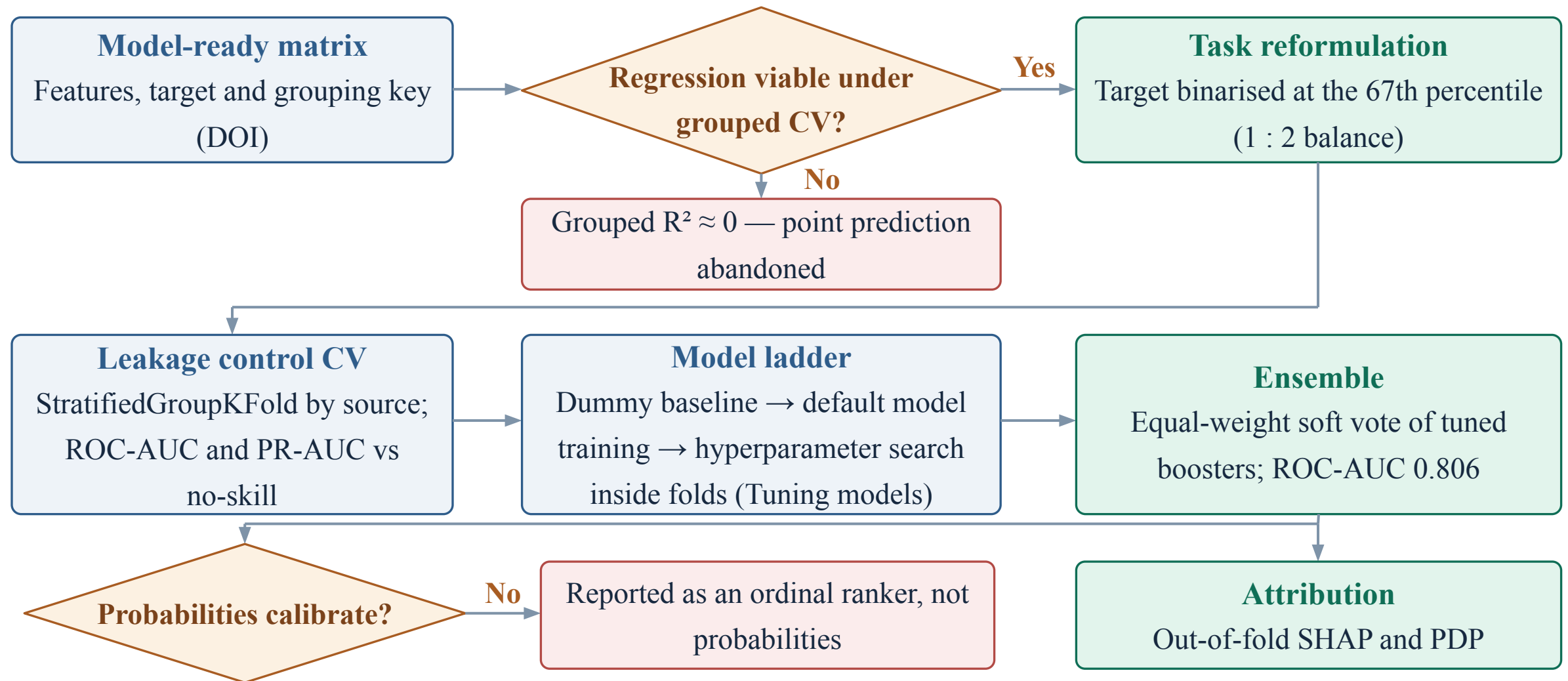
0.566 ROC confirms a nonlinear decision boundary.

Trees converge

0.75–0.80 across four algorithms — means a data ceiling, not luck.

Modelling & XAI

Flowchart for Modelling and explainability



Model performances

Model	ROC-AUC	PR-AUC
Dummy (stratified)	0.476	0.324
Logistic regression	0.566	0.476
Random forest	0.753	0.668
XGBoost	0.762	0.671
HistGradientBoosting	0.766	0.656
CatBoost	0.798	0.636
Soft-vote ensemble (Cat+Hist+XGB)	0.806	0.661

ROC-AUC and PR-AUC under paper-grouped 5-fold CV (StratifiedGroupKFold). No-skill: 0.500 / 0.333.

Picked models

- **Built a classification-oriented ML model** that ranks compositions by their propensity to reach top-tier specific capacitance (top-33%, $C_{sp} > 339.7$ F/g).

Chosen model ensemble:

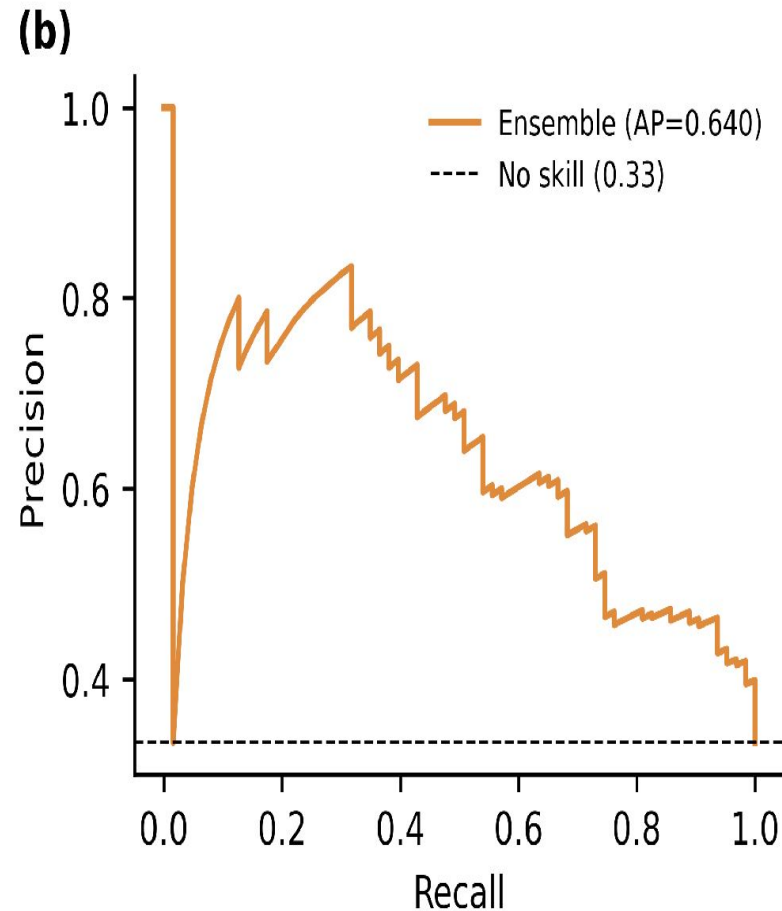
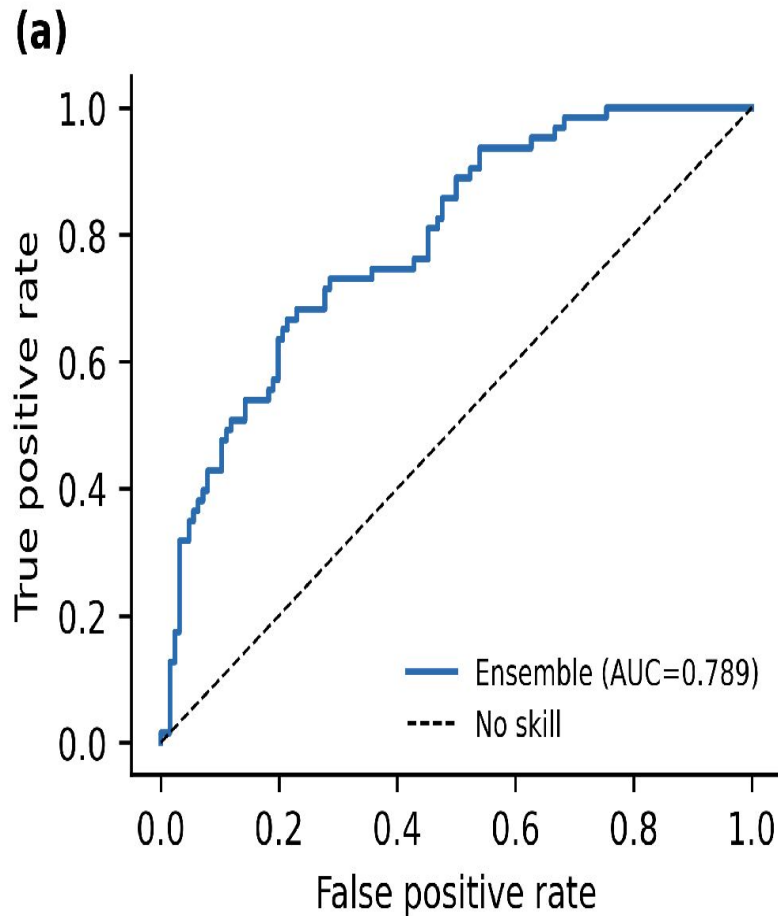
Ensemble	ROC-AUC	PR-AUC
Cat + Hist + XGB (equal weighted)*	0.806	0.661
Cat + Hist	0.807	0.652

***Chosen:** 3-way diversity gives the best PR-AUC and most robust ranking. Progression: 0.762 (best single model) → 0.806 (ensemble).

Validated with paper-grouped 5-fold CV (no source-paper leakage); no-skill floor at $ROC = 0.500$.

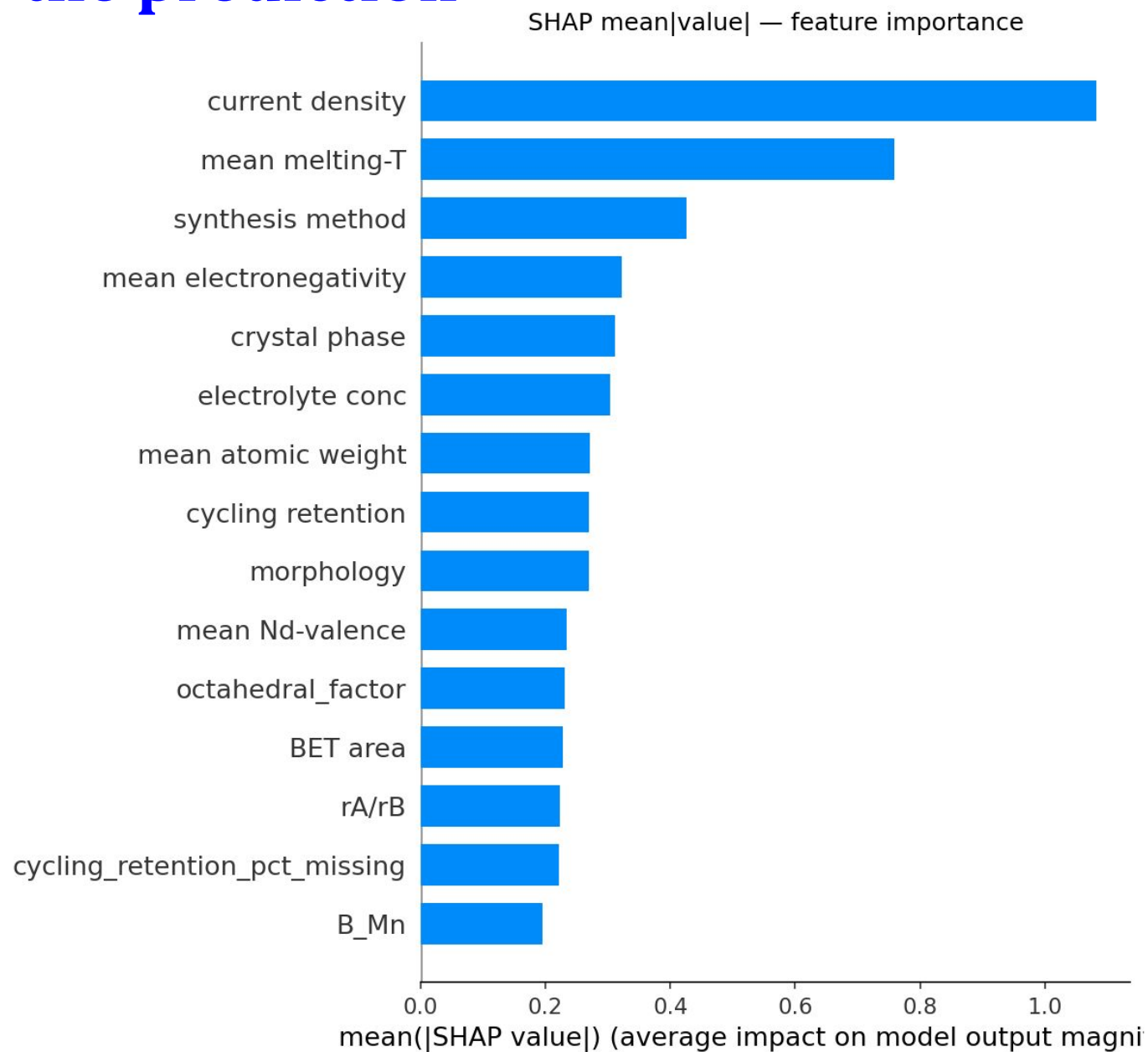
The delivered model: equal-weight soft vote

CatBoost + HistGB + XGBoost — ROC-AUC 0.806, PR-AUC 0.661

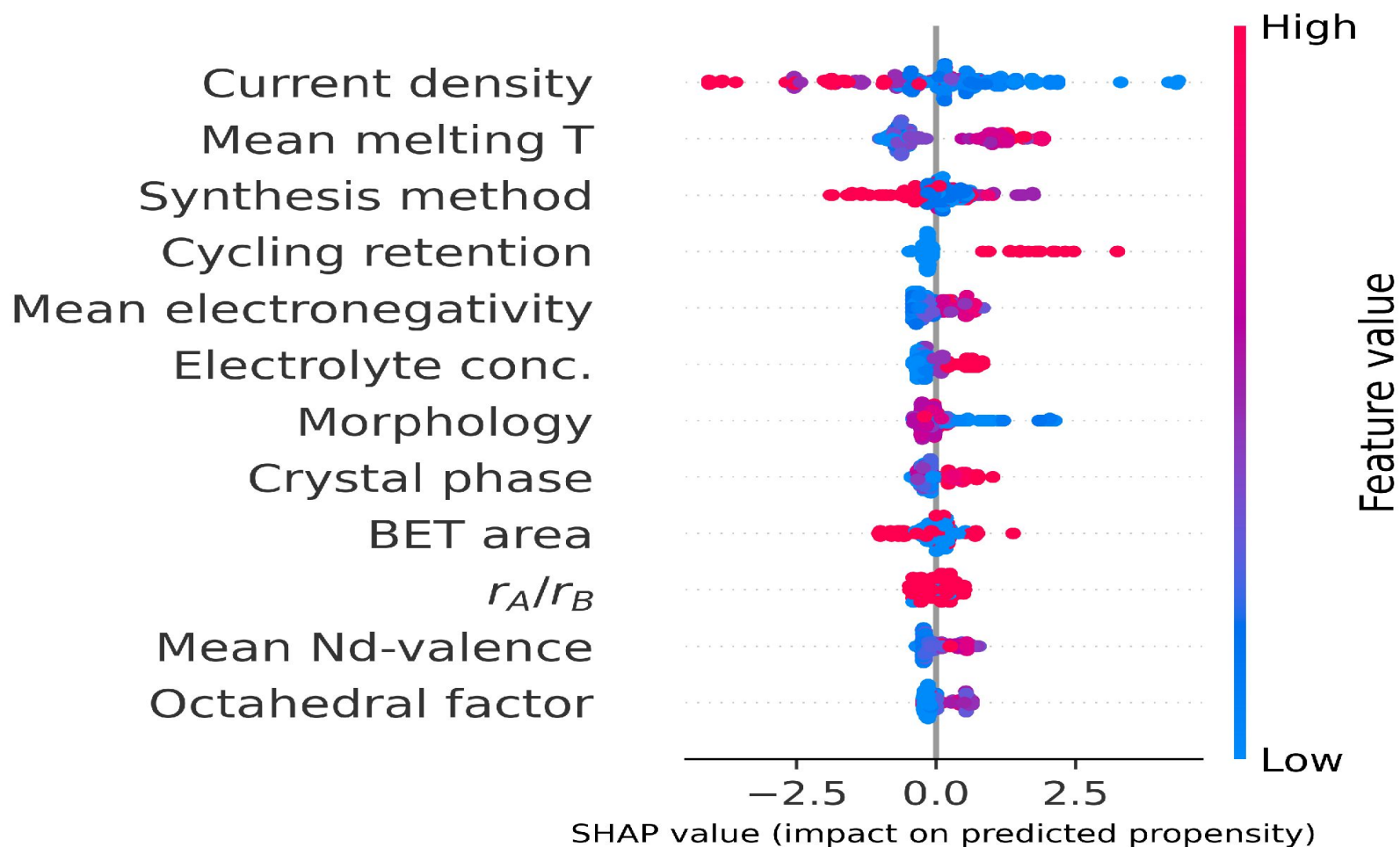


- **Tuning: marginal**
0.762 → 0.783 best single. The framing did the work, not the knobs.
- **Ensembling: real**
The three make different errors, so voting genuinely helps: 0.806.
- **Honest ceiling**
Three independent boosters agreeing at ≈ 0.79 is the true signal strength.

What drives the prediction



What drives the prediction



● Melting T $\uparrow \rightarrow$ better

Strong metal–oxygen bonding, robust redox-active lattice.

● Current density $\uparrow \rightarrow$ worse

Classic rate limitation — basic electrochemistry, recovered.

● Cross-check

Two independent modellings (earlier regression & now classification) agree on drivers

Explainability XAI

→ **Applied SHAP explainability** to the ensemble (out-of-fold) to identify which features drive high-Csp propensity.

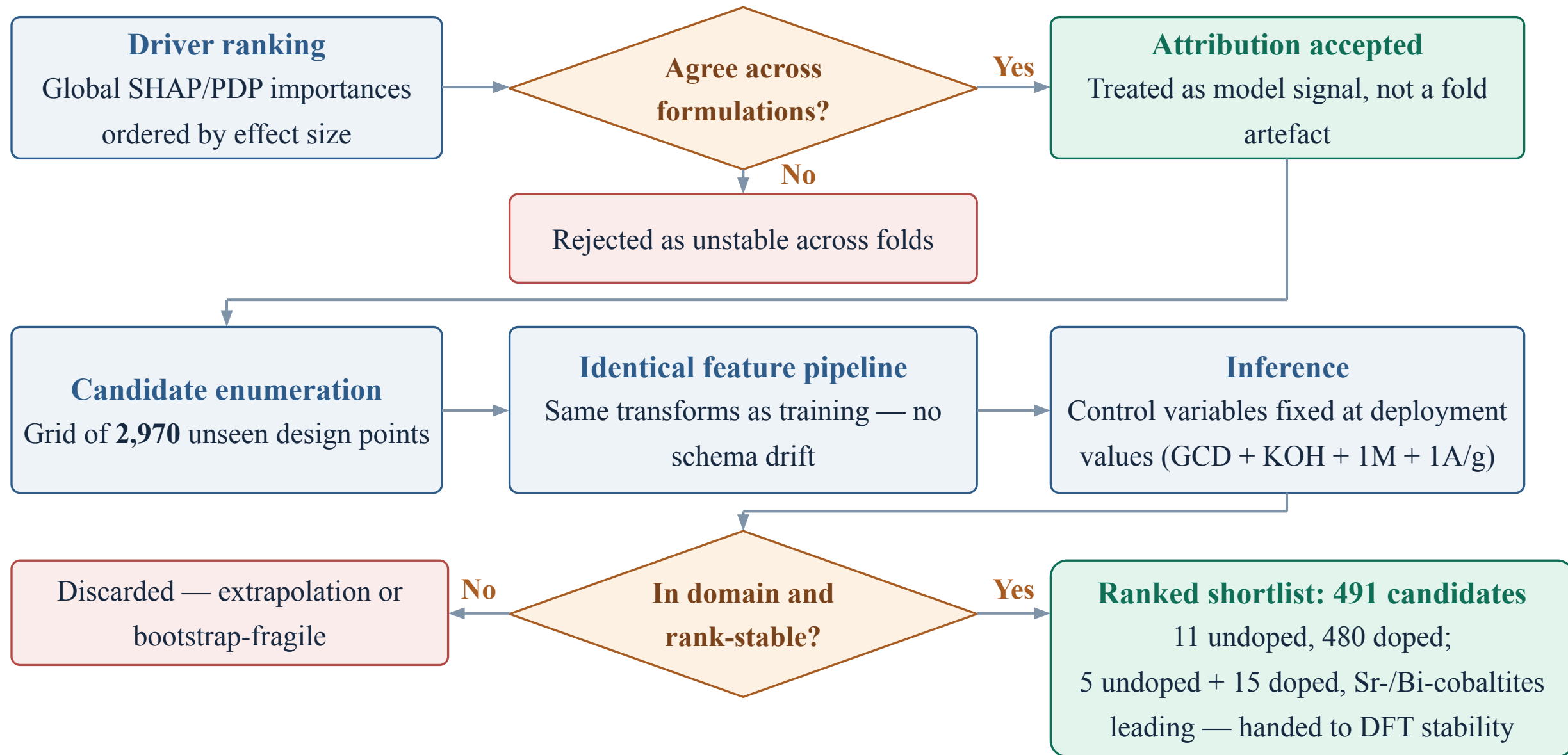
Top drivers among several others:

Feature	Family	Effect on high-Csp likelihood
mean melting-T	Chemistry	Higher → more likely (robust M–O bonding / redox lattice)
current density	Condition	Higher → less likely (rate limitation)
synthesis method	Processing	Governs morphology & surface area
mean electronegativity	Chemistry	Bonding character / redox activity
crystal phase	Structure	Structural family of the composition

These drivers reappeared independently in the earlier regression work → the signal is real chemistry, not an artifact.

Ranking Candidates

Flowchart for Interpretation, screening and ranking



Screening 2,970 candidates

Why this many:

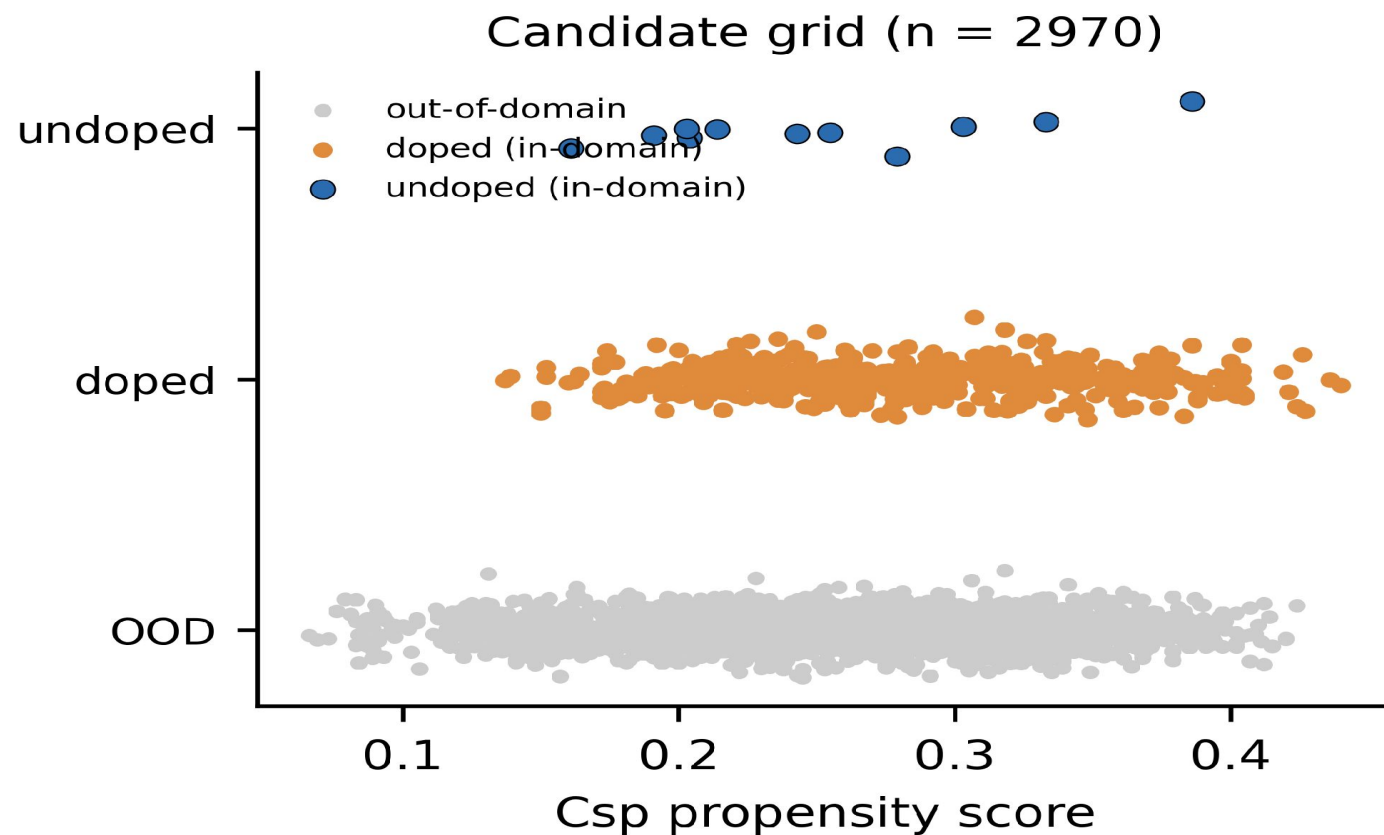
- **10 A-site cations** — La, Sr, Ca, Ba, Gd, Nd, Pr, Y, Bi, K
- **9 B-site cations** — Mn, Co, Fe, Ni, Cr, Ti, V, Nb, Ru

Undoped: every A×B pairing $\rightarrow 10 \times 9 = \mathbf{90}$ compositions (LaMnO₃, SrCoO₃, BiNiO₃...).

Doped: for each of those 90 hosts, pick a *different* B-site element as dopant (8 remaining choices), at 4 fractions ($x = 0.1, 0.2, 0.3, 0.5$) $\rightarrow 90 \times 8 \times 4 = \mathbf{2,880}$ (e.g. SrCo_{0.9}Ni_{0.1}O₃).

Total: $90 + 2,880 = \mathbf{2,970}$.

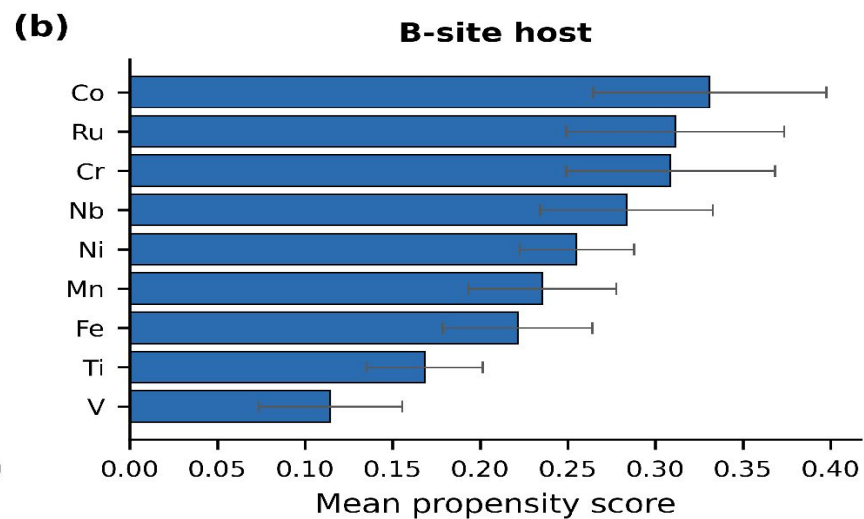
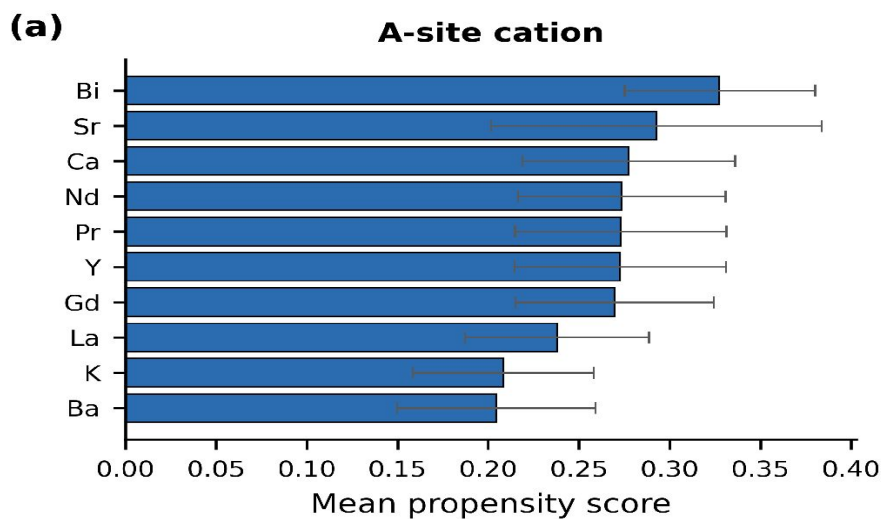
Screening 2,970 candidates



- Featurised properly**
matminer Magpie + Shannon-radii
tolerance/octahedral factors for novel
compositions.
- Rank stability**
30 bootstrap refits give every
candidate a score SD.
- Domain flag**
Chemistry-space only — guards
against confident-looking
extrapolation.

Score distribution by class and applicability domain. 491 of 2,970 are chemistry-in-domain.

Which lever actually matters



→ **B-site host dominates**

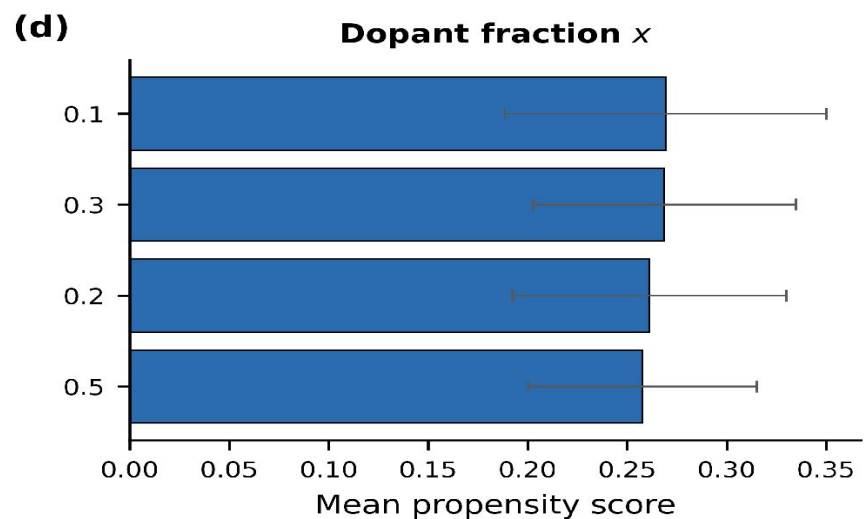
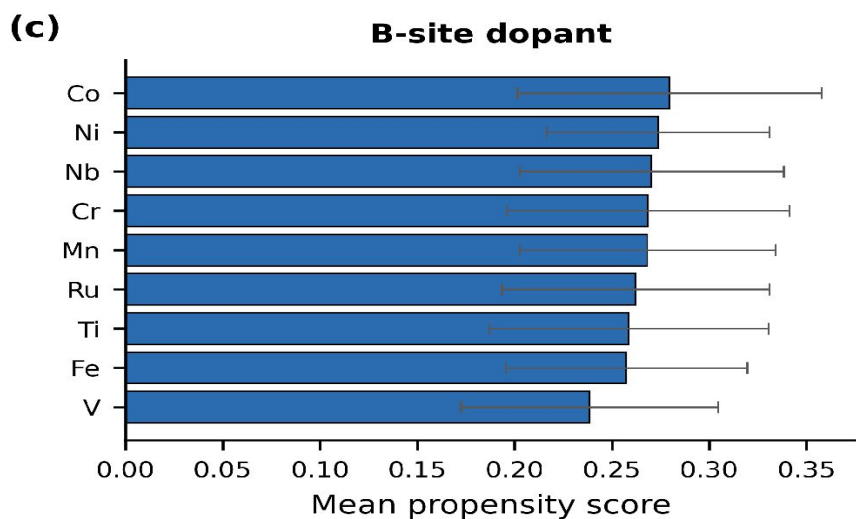
Range 0.11–0.33. Co → Ru → Cr. The strongest, most trustworthy lever.

→ **A-site second**

Bi and Sr lead; Ba and K consistently worst.

→ **Doping fine-tunes**

Dopant identity weak (0.24–0.28); fraction x flat, within noise.



Hand-off Candidates for Stability Screening

→ **Top-ranked chemistry-in-domain candidates handed to DFT validation** (scored at fixed conditions: 1 A/g, KOH, GCD, 1 M).

Top 5 Undoped Candidates

Composition	pctile	$\pm$ SD	risk	DFT
SrCoO ₃	99	0.24	support.	P1
LaCoO ₃	83	0.23	support.	P1
LaCrO ₃	73	0.23	support.	P1
SrMnO ₃	62	0.23	support.	P1
GdNiO ₃	54	0.21	support.	P1

All P1 (data-supported); genuine score gradient (0.39 → 0.26).

Top 5 Doped Candidates

Composition	x	pctile	$\pm$ SD	risk	DFT
BiCo _{0.9} Ni _{0.1} O ₃	0.1	100	0.22	extrap.	P2
SrCo _{0.9} Ni _{0.1} O ₃	0.1	100	0.24	support.	P1
BiCr _{0.9} Co _{0.1} O ₃	0.1	100	0.22	extrap.	P2
SrCr _{0.9} Co _{0.1} O ₃	0.1	100	0.24	extrap.	P2
CaCo _{0.9} Ni _{0.1} O ₃	0.1	100	0.23	extrap.	P2

Scores within one $\pm$ SD ($\approx$ 0.22) → statistically tied; select by risk / DFT priority.

P1 = data-supported, validate first **P2** = extrapolated, DFT checks if real • score = ordinal Csp-propensity (not a probability)

Stability Screening

Three questions, Three targets, Four models

Why formation energy and band gap — and where the stability percentage comes from

Question	Target	Model	Role
Is the compound energetically favourable?	Formation energy (eV/atom)	XGBoost regression	First-order stability indicator — how strongly A + B + O want to combine
Will it conduct as an electrode?	Band gap (eV)	Two-stage: metal classifier → gap regression	Electrodes need gap ≈ 0 ; conductivity screen
What is the chance it is synthesizable?	Stable / unstable label ($E_{\text{hull}} \leq 50$ meV/atom)	XGBoost + isotonic calibration	The stability percentage — a real, calibrated probability

The dataset — Materials Project DFT records

One clean, single-source database — the opposite of the literature-mined C² data

2,563 entries

raw MP query — anonymous formula ABC₃
containing O

1,319 compositions

strict ABO₃ (1:1:3), lowest-energy polymorph kept

1,000 rows modelled

carbonates / nitrates / selenites removed (319 impostors)

What each row carries

Targets

formation energy/atom · band gap (+ E_{hull} kept for the stability label)

Site assignment

A vs B decided by electronegativity — MP returns elements unordered

Features (103)

identical stack to Stage 1: Magpie statistics + Shannon-radii tolerance & octahedral factors + r_A/r_B, with missingness flags

Composition split

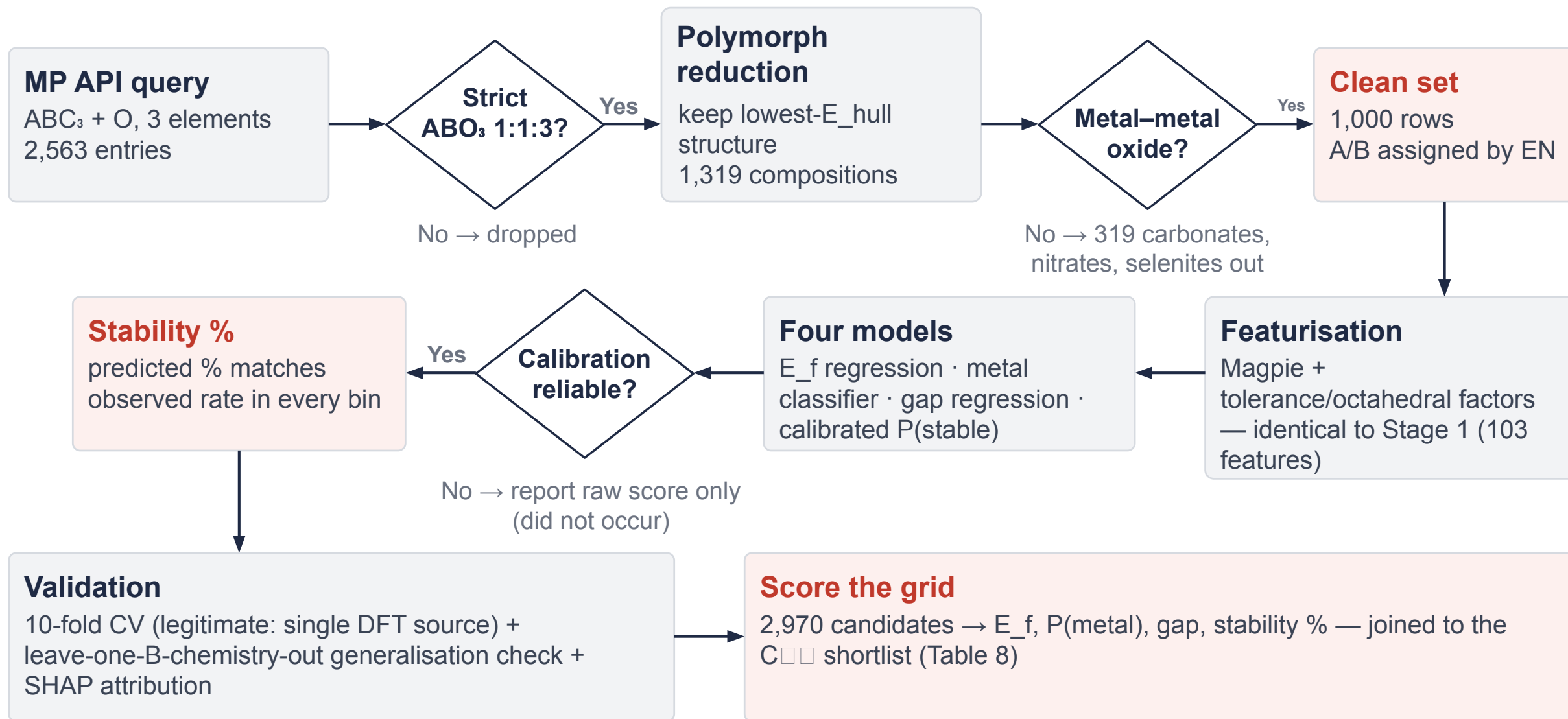
646 metals (gap = 0) vs 354 gapped → justifies the two-stage gap model

Ground truth link

259 rows matched to experimental (ICSD) structures

All 36 elements of the screening grid appear in this dataset — the stability model has direct support across the entire candidate space.

Flowchart on Data Acquisition, Cleaning and Modelling



Model performance

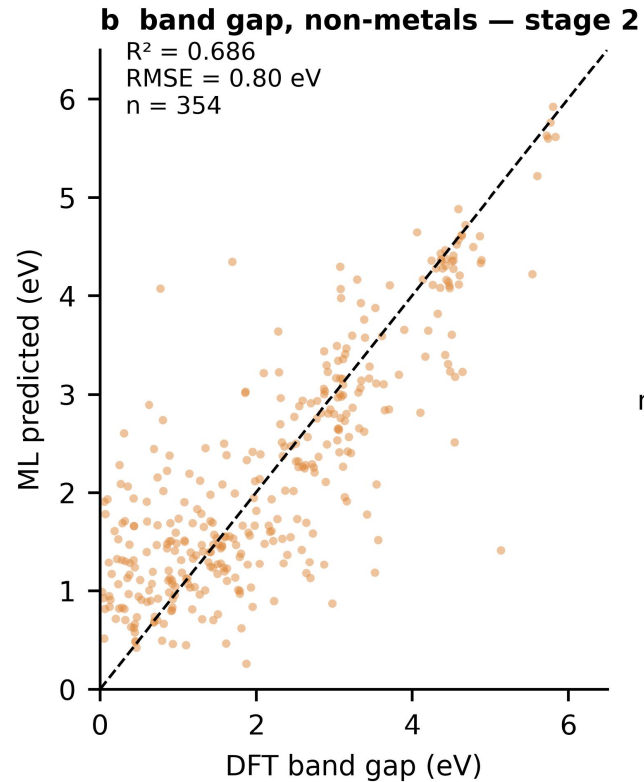
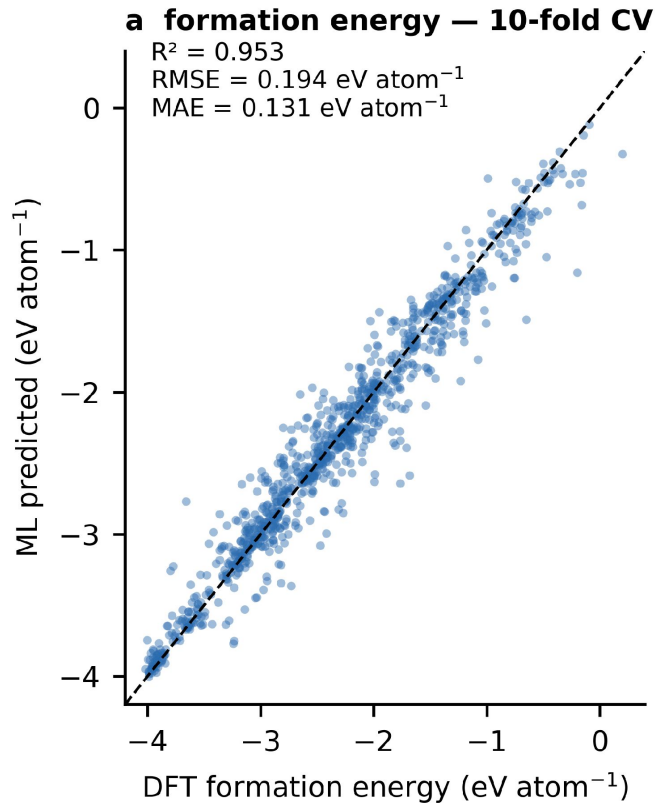
Model performances

10-fold cross-validation on 1,000 Materials Project compositions

Model	Task	Key metrics
XGBoost regression	Formation energy	R ² 0.953 · RMSE 0.194 · MAE 0.131 eV/atom
HistGradientBoosting (check)	Formation energy	R ² 0.950 · RMSE 0.199 eV/atom — confirms XGB
XGBoost classifier — stage 1	Metal vs non-metal	ROC-AUC 0.899 · accuracy 0.82 · F1 0.86
XGBoost regression — stage 2	Band gap (354 non-metals)	R ² 0.686 · RMSE 0.80 eV — screen-grade, not spectroscopy
XGBoost + isotonic calibration	P(stable), E_hull ≤ 50 meV label	ROC-AUC 0.91 · accuracy 0.83 · Brier 0.119

Predicted vs DFT — the two regression targets

Out-of-fold predictions; every point is unseen by the model that scored it



Formation energy: tight

R^2 0.953 across a 4 eV/atom span.
Composition alone nearly determines E_f — expected, and it transfers to doped grid compositions.

Band gap: usable

Scatter grows above $\sim 3 \text{ eV}$, but the screen only asks conductor vs insulator — and stage 1 answers that at ROC-AUC 0.90.

Known caveat

DFT (GGA) underestimates gaps systematically — fine for a 0-vs-nonzero screen, stated in limitations.

The stability percentage is a real probability

Isotonic-calibrated classifier on the $E_{\text{hull}} \leq 50$ meV/atom label — reliability verified out-of-fold

Model says	Actually stable	n
0–10 %	3.0 %	165
30–40 %	33.3 %	60
60–70 %	65.2 %	69
80–90 %	93.8 %	80
90–100 %	94.4 %	72

What calibration means

When the model says 65 %, about 65 % of such compounds really are stable — in every bin, across the whole range. The percentage can be quoted literally.

The contrast from earlier C_{sp} model

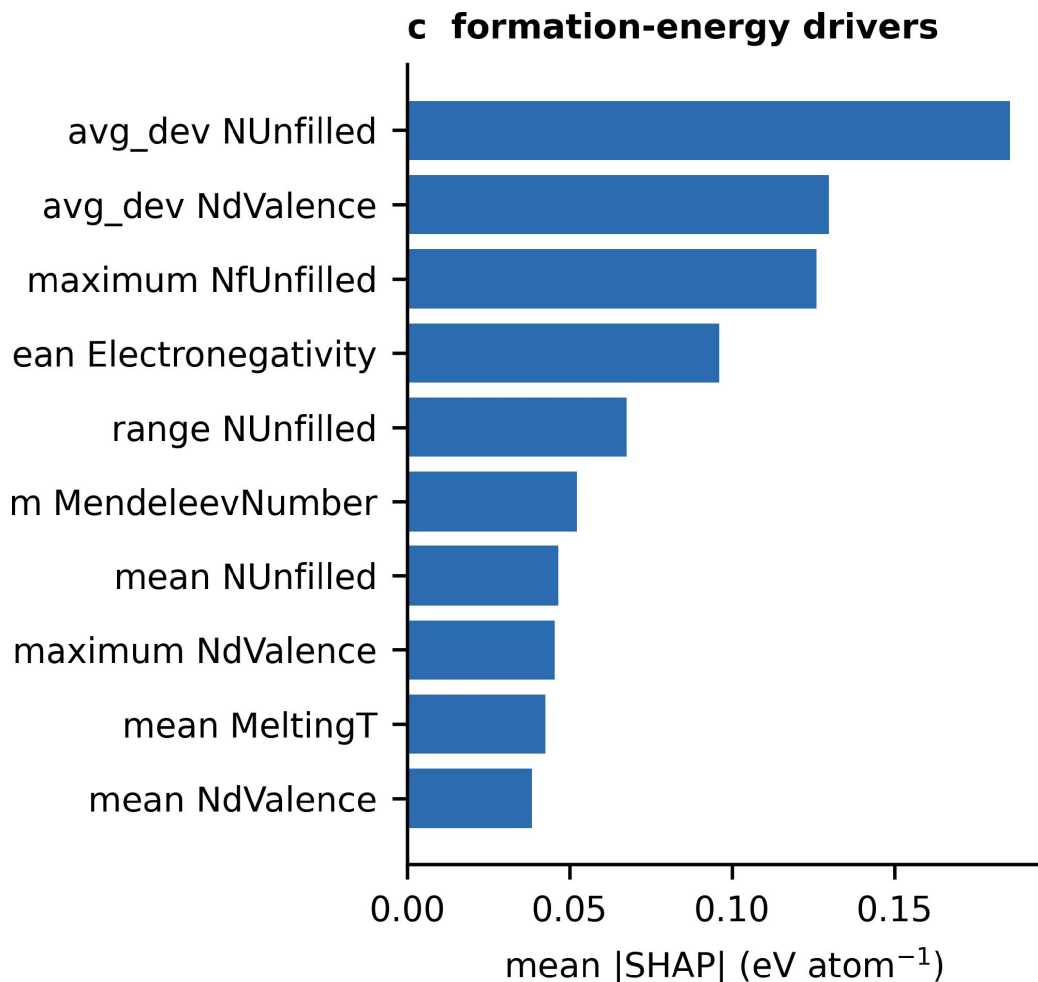
Calibration FAILED for the $C_{\square\square}$ model (29 source papers — too few groups). Here it works: 1,000 rows, one consistent DFT source. Failure and success have the same cause: data structure.

So the outputs differ on purpose

$C_{\square\square}$ model → ordinal ranking score ("ranks above 95 %"). Stability model → calibrated percentage ("89 % likely stable").

What drives predicted stability

SHAP on the formation-energy model — and it is textbook oxide chemistry



Valence-electron structure leads

Unfilled d/f states (avg dev NUnfilled, NdValence) dominate — how many bonding electrons the cations can commit to the M–O framework.

Electronegativity second

Mean electronegativity sets the ionic/covalent character of the M–O bonds — the classic thermochemical driver of oxide formation.

Echoes earlier C_{sp} Model

Electronegativity and melting-T-family features drive both models — the C_{sp} signal and the stability signal share chemical roots.

Results

The Stability Screen separates the two risk tiers

Candidate (C□□ rank tier)	E_f (eV/atom)	Stability %	Predicted gap
SrCo_{0.9}Ni_{0.1}O₃ — P1	−1.94	92 %	metal (0 eV)
SrCoO₃ — P1	−1.98	90 %	metal (0 eV)
SrMnO₃ — P1	−2.48	94 %	metal (0 eV)
LaCoO₃ — P1	−2.61	91 %	0.28 eV
BiCo_{0.9}Ni_{0.1}O₃ — P2	−1.35	49 %	metal (0 eV)
CaCo_{0.9}Ni_{0.1}O₃ — P2	−1.93	32 %	metal (0 eV)
BiCr_{0.9}Co_{0.1}O₃ — P2	−1.77	51 %	1.32 eV

Thank You